

Data Processing

Learning Objectives

- Process collections of data using iteration, filtering, and aggregation
- Apply statistical operations to datasets using appropriate data structures
- Transform raw data into meaningful information through structured processing

Engage (5-7 minutes)

Raw data is like unrefined crude oil—valuable but useless until processed. A weather station collects thousands of temperature readings, but what matters is the daily average, the monthly trend, and the anomalies. Data processing transforms raw information into actionable insights. Today we learn to extract meaning from data using the data structures we have mastered.

Explore (10-12 minutes)

Given this dataset of 20 student records (name, age, marks in 3 subjects), write code to: calculate the class average per subject, find the student with the highest total marks, identify students who need remedial support (below 40 in any subject), and generate a summary report. Think about which data structures work best for each task. Discuss: should you process data in-place or create new structures?

Explain (10-12 minutes)

Data processing involves three core operations: filtering (selecting relevant records), transformation (converting data to desired format), and aggregation (computing summaries). Common aggregations include sum, count, average, min, max, and standard deviation. Processing can be done procedurally (step-by-step loops), functionally (using map, filter, reduce), or with comprehensions. Understanding data flow—how data moves through each transformation step—is crucial for writing efficient and correct processing pipelines.

Elaborate (8-10 minutes)

Process this dataset: a CSV-like list of 50 sales records with date, product, quantity, and price. Write functions to: calculate total revenue per product, find the best-selling day, compute monthly sales trends, identify products that declined in sales, and generate a formatted report. Use appropriate data structures for each intermediate result. Then optimize your code by combining multiple passes through the data into a single loop. Discuss memory efficiency when processing large datasets.

Evaluate (5-8 minutes)

Given a list of dictionaries representing exam results, write a complete data processing pipeline that: filters passing students (average ≥ 40), calculates grade distribution (A, B, C, D, F), identifies the top 3 performers, and generates a summary dictionary with all statistics. Show the data structure at each processing step.